Learning Dataset-Invariant Tactile Latents from Egocentric Video

Project Proposal

Codrin Crismariu
Mechanical Engineering
Tufts University

August 24, 2026

Abstract

Touch provides information about contact, pressure, force, slip, and compliance that is difficult or impossible to measure from appearance alone. However, large-scale human and robot manipulation datasets are dominated by egocentric video, while the smaller datasets that include tactile measurements use incompatible sensors and representations. This project asks whether heterogeneous vision–tactile datasets can supervise a shared latent representation of touch predicted from egocentric video. The initial goal is to train a common visual encoder whose tactile latent supports multiple dataset- and sensor-specific tactile decoders, while preventing the latent from merely encoding dataset identity. The project will evaluate whether the resulting representation is dataset-invariant, physically meaningful, and suitable as a reusable tactile prior across heterogeneous training sources. Downstream outcomes are intentionally left broad: promising representations may support tactile prediction, contact-risk estimation, uncertainty-aware manipulation, or tactile-token augmentation of vision-language-action policies.

1 Motivation

Vision-language-action (VLA) models benefit from abundant visual and linguistic supervision, but contact-rich manipulation depends on physical properties that are poorly observed through an external camera. Direct tactile sensing can provide these properties, yet tactile hardware is heterogeneous and tactile datasets remain fragmented across optical sensors, pressure arrays, wearable gloves, and force–torque measurements. Consequently, a model trained to predict one sensor's output may learn that sensor's conventions rather than a transferable concept of touch.

This project explores a complementary direction: learning a normalized tactile latent from human and robot egocentric video using heterogeneous tactile supervision. Instead of requiring every dataset to share a sensor or output representation, a single visual encoder will produce a shared tactile latent, and sensor-specific decoders will reconstruct or predict the tactile targets available in each dataset. If the latent captures common contact phenomena rather than dataset identity, it could provide a reusable physical prior for manipulation models even when a particular tactile sensor is unavailable.

The project aligns with SPARC Lab's goal of safe, deployable, and trustworthy robot autonomy. In particular, it addresses learned perception under heterogeneous sensing and distribution shift. A successful representation could later support uncertainty-aware contact-risk estimation or conservative manipulation decisions, while the initial project remains focused on representation learning and validation.

2 Research Question and Hypotheses

The central research question is:

> Can heterogeneous human and robot vision–tactile datasets supervise a shared latent representation of touch that is predictable from egocentric video, invariant to dataset identity, and transferable across sensors and embodiments?

The initial study will test four hypotheses:

1. Joint multi-dataset training produces a representation that transfers better than independently trained, dataset-specific predictors.
2. Reconstruction through multiple sensor-specific decoders is not sufficient by itself to produce invariance; explicit domain-invariance and shared-physics objectives improve transfer.
3. A short temporal window is more informative than a single frame for representing contact onset, deformation, and changing force.
4. The tactile latent supports physical probes shared across the heterogeneous training datasets.

3 Relationship to Prior Work

Recent research establishes both the opportunity and the need for this study. EgoTac predicts continuous force and binary contact from egocentric video using a large unified collection of image–tactile pairs [10]. TouchAnything introduces a bimanual dataset and predicts tactile pressure from egocentric and wrist-mounted video [11]. Imagine2Touch predicts expected tactile measurements from a visual patch for downstream object recognition [1]. These projects demonstrate that vision can supervise useful tactile predictions, but they do not establish a general, sensor-independent tactile latent intended to transfer across heterogeneous tactile targets and downstream models.

A parallel literature learns representations directly from tactile measurements. UniTouch uses sensor-specific tokens to train a unified model across vision-based tactile sensors and aligns touch with vision, language, and sound [8]. Sparsh learns general-purpose self-supervised representations for several vision-based tactile sensors [5]. AnyTouch learns unified static and dynamic representations across multiple optical tactile sensors [3], while Sensor-Invariant Tactile Representation studies zero-shot transfer across optical sensor designs [4]. These methods primarily encode tactile observations rather than infer a common tactile latent from ordinary egocentric video.

FTP-1 aggregates human and robot demonstrations across heterogeneous tactile hardware and maps them into morphology-aware tokens for a generalist tactile policy [9]. Its representation is learned inside a $\pi_{0.5}$-based action policy and requires tactile measurements as inputs. In contrast, the proposed project focuses on a policy-independent latent predicted from egocentric visual observations. The representation may later augment a VLA, but action modeling is not required for the initial study.

4 Proposed Method

Let dataset $d \in \{1, \ldots, D\}$ contain an egocentric video window $I^{(d)}_{t-k:t}$ and tactile target $x^{(d)}_t$. Tactile targets may be pressure maps, force vectors, binary contacts, or optical tactile observations. A shared visual-temporal encoder E_θ produces a latent distribution

$$q_\theta(z_t \mid I^{(d)}_{t-k:t}) = \mathcal{N}\big(\mu_t, \mathrm{diag}(\sigma_t^2)\big)\,. \tag{1}$$

Each dataset or sensor has a decoder D_{ϕ_d} appropriate to its output type:

$$\hat{x}^{(d)}_t = D_{\phi_d}(z_t). \tag{2}$$

Only the decoder corresponding to the available target is evaluated for a training example. Candidate decoder losses include regression for force and pressure, classification for contact, and perceptual or reconstruction objectives for optical tactile images.

4.1 Candidate Visual Encoders

Several encoder architectures will be tested rather than selecting one in advance. DINO and DINOv2 are promising options because their self-supervised features transfer well across image datasets and preserve spatial information that may help localize contact [2, 6]. However, they process images independently, so they may miss changes in force or contact that are only visible over time. A standard CNN or Vision Transformer may be easier to train and adapt, but may also learn dataset-specific visual shortcuts more readily.

Temporal models will also be considered. One option is to combine frame-level features with a small temporal transformer or recurrent network. Another is a video encoder such as VideoMAE, which models space and time jointly but is more computationally expensive [7]. The experiments will compare multiple encoder families, training strategies, and global versus spatial features to determine which produces the most physically meaningful and dataset-invariant tactile latent.

4.2 Preventing Dataset Collapse

A shared tactile latent does not automatically become a normalized representation. The encoder could retain camera style, task identity, or other dataset-specific shortcuts that allow each decoder to solve its own task. The project will compare several mechanisms for discouraging this failure:

- **Capacity control:** constrain the dimensionality or information capacity of the tactile latent.
- **Domain-adversarial learning:** train a classifier to identify the source dataset from z, while reversing its gradient with respect to the encoder.
- **Shared physical heads:** predict standardized properties such as contact presence, contact region, or normalized force wherever compatible labels can be derived.
- **Cross-dataset alignment:** bring physically similar interactions together using labels, temporal events, or weak semantic descriptions while avoiding indiscriminate distribution matching.
- **Appearance randomization:** reduce shortcuts caused by camera color, resolution, background, embodiment, and dataset-specific preprocessing.

A general objective is

$$\mathcal{L} = \sum_d \mathcal{L}^{(d)}_{\text{decode}} + \lambda_{\text{phys}}\mathcal{L}_{\text{shared-physics}} + \lambda_{\text{align}}\mathcal{L}_{\text{alignment}} - \lambda_{\text{domain}}\mathcal{L}_{\text{dataset-classifier}} + \beta\mathcal{L}_{\text{latent-regularization}}. \tag{3}$$

The sign on the domain-classification term reflects adversarial optimization: the classifier minimizes its error, while the encoder is trained to make dataset identity difficult to recover.

5 Evaluation Plan

Evaluation will separate output prediction from representation quality. This is essential because accurate in-domain decoding does not prove that the tactile latent is normalized or transferable.

5.1 In-Domain Prediction

For each dataset, the proposed model will be compared with an independently trained dataset-specific predictor. Metrics will match the output modality, including force error, pressure-map error, contact F1 or intersection-over-union, and temporal correlation with measured tactile signals.

5.2 Dataset Invariance

Dataset identity will be predicted from a frozen latent using a probe. High tactile-task accuracy combined with low dataset-classification accuracy would provide evidence that the representation retains physical information while suppressing domain shortcuts. Latent visualizations will be used only as supporting evidence, not as the primary invariance metric.

5.3 Cross-Dataset Physical Structure

Linear probes and retrieval experiments will test whether contact state, force range, temporal phase, and interaction type organize the latent consistently across datasets. Where sufficiently compatible labels exist, an observation from one dataset will be used to retrieve physically similar interactions from another.

5.4 Uncertainty and Downstream Utility

If the probabilistic latent is well calibrated, uncertainty will be evaluated under visual occlusion, unfamiliar objects, and shifts in camera or environment appearance. Depending on initial results and available hardware, a downstream study may investigate one of the following:

- calibrated bounds on a safety-relevant quantity such as contact force;
- detection of interactions for which visually inferred touch is unreliable;
- augmentation of a VLA or imitation-learning policy with predicted tactile tokens; or
- safety-aware manipulation that slows, stops, or requests physical tactile feedback under high uncertainty.

These are candidate outcomes rather than commitments of the initial representation-learning phase.

6 Expected Contributions

The anticipated contributions are:

1. A unified formulation for learning an egocentric tactile latent from heterogeneous human and robot datasets with incompatible tactile targets.

2. An empirical study of when multi-decoder learning produces a shared physical representation and when it collapses to dataset-specific shortcuts.

3. A benchmark protocol based on dataset probing, cross-dataset physical retrieval, shared physical-property probes, and controlled comparisons of visual encoder families.

4. A reusable tactile-prior representation that may support uncertainty estimation, manipulation policies, or VLA tactile tokens.

7 Risks and Mitigations

Visual ambiguity. Force, friction, hidden contact, and internal grasp pressure are not uniquely identifiable from video. The model will represent uncertainty rather than claim that predicted touch is equivalent to physical sensing.

Incompatible labels. Datasets differ in calibration, sampling rate, spatial layout, and semantics. The first experiments will use a small number of compatible targets and expand only after establishing reliable synchronization and normalization.

Dataset shortcuts. The encoder may identify the dataset from visual style. Strong augmentations, adversarial probes, cross-dataset evaluation, and explicit ablations will test and mitigate this behavior.

Negative transfer. Not all tactile targets share the same information. The project will compare fully shared, partially shared, and grouped latent structures rather than assume complete universality.

8 Provisional Timeline

Phase	Activities
1	Select datasets; document sensor formats, calibration, temporal alignment, licenses, and compatible targets.
2	Implement dataset-specific baselines and a shared encoder with multiple decoders.
3	Add latent regularization, domain-adversarial, and shared-physics objectives; conduct controlled ablations.
4	Evaluate dataset identifiability, latent probes, cross-dataset retrieval, and DINO/DINOv2 temporal-encoder variants.
5	Select a downstream direction based on observed latent properties and available hardware.
6	Complete experimental analysis, limitations, and final paper.

9 Conclusion

This project investigates whether heterogeneous tactile supervision can teach a shared visual model a transferable representation of touch. Its initial objective is deliberately foundational: construct

and rigorously evaluate a normalized tactile latent over human and robot egocentric video without allowing the representation to reduce to dataset identity. Rather than predetermining a single application, the project will use the learned representation's measured properties to motivate an appropriate downstream outcome, potentially including calibrated contact-risk estimation, uncertainty-aware manipulation, or tactile augmentation of VLA policies.

References

[1] Abdallah Ayad, Adrian Röfer, Nick Heppert, and Abhinav Valada. Imagine2touch: Predictive tactile sensing for robotic manipulation. *arXiv preprint arXiv:2405.01192*, 2024. URL `https://arxiv.org/abs/2405.01192`.

[2] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021. URL `https://arxiv.org/abs/2104.14294`.

[3] Ruoxuan Feng, Jiangyu Hu, Wenke Xia, et al. Anytouch: Learning unified static-dynamic representation across multiple visuo-tactile sensors. In *International Conference on Learning Representations*, 2025. URL `https://openreview.net/forum?id=XToAemis1h`.

[4] Harsh Gupta, Yuchen Mo, Shengmiao Jin, and Wenzhen Yuan. Sensor-invariant tactile representation. In *International Conference on Learning Representations*, 2025. URL `https://arxiv.org/abs/2502.19638`.

[5] Carolina Higuera, Akash Sharma, Chaithanya Krishna Bodduluri, et al. Sparsh: Self-supervised touch representations for vision-based tactile sensing. *arXiv preprint arXiv:2410.24090*, 2024. URL `https://arxiv.org/abs/2410.24090`.

[6] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. URL `https://arxiv.org/abs/2304.07193`.

[7] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training. In *Advances in Neural Information Processing Systems*, 2022. URL `https://arxiv.org/abs/2203.12602`.

[8] Fengyu Yang, Chao Feng, Ziyang Chen, Hyoungseob Park, Daniel Wang, Yiming Dou, Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, and Alex Wong. Binding touch to everything: Learning unified multimodal tactile representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. URL `https://arxiv.org/abs/2401.18084`.

[9] Chengbo Yuan, Zicheng Zhang, Mingjie Zhou, et al. Ftp-1: A generalist foundation tactile policy across tactile sensors for contact-rich manipulation. *arXiv preprint arXiv:2606.13102*, 2026. URL `https://arxiv.org/abs/2606.13102`.

[10] Wenkang Zhang, Chengbo Yuan, Zicheng Zhang, Zhengxue Cheng, and Yang Gao. Egotac: In-the-wild tactile prediction from egocentric vision. *arXiv preprint arXiv:2608.15060*, 2026. URL `https://arxiv.org/abs/2608.15060`.

[11] Jianyi Zhou, Ziteng Gao, Feiyang Hong, et al. Touchanything: A dataset and framework for bimanual tactile estimation from egocentric video. *arXiv preprint arXiv:2605.13083*, 2026. URL `https://arxiv.org/abs/2605.13083`.